

SCOUT → COMPASS: Handoff Report (Basketball / Sports Domain)

Rename (2026-06-11): Formerly `SCOUT_report_to_master_planner.md`; planning agent is **COMPASS**. See `COMPASS_AGENT_IDENTITY.md`.

Agent: SCOUT (Cursor agent; NCAA basketball panel, Tier 1 modeling, generative simulation, HPC sweeps)

Domain root: `./sports/` (panel + Tier 1 modules; shared panel CSV at workspace `datasets/mbb/`)

Report date: 2026-06-08

Prepared for: Charles Levine → future **COMPASS** agent (Cursor)

Charles's explicit instruction (2026-06-08): This report feeds the COMPASS. **Do not** begin drafting the cross-project master plan until Charles **assigns** that agent and asks for planning work. SCOUT/PEER/CODA/VECTOR should only **report status** until then.

Companion reports (same folder): - `CODA_report_to_COMPASS.md` — Army / `talent/` - `PEER_report_to_COMPASS.md` — Academia / `tenure/` - **VECTOR report:** Not yet in repo as of this writing — expect manuscript-focused handoff from Scholar GPT workflow or a Cursor session dedicated to `5-Manuscript/`.

0. What this document is (and is not)

What it is

- A **ground-truth status snapshot** of the basketball/sports lane as SCOUT understands it after work through **June 2026** (530 inverted-U replicated; 538 empirical Wang ladder; 538D generative CELL 10 congestion score; axis-conditioning finding; documentation pass).
- A **file map** so the COMPASS can navigate without re-deriving context from chat logs, SpecStory exports, or stale May briefs.
- A **list of open questions** where Charles's intent is not fully locked — SCOUT prefers explicit answers over assumptions.
- Pointers to **CODA**, **PEER**, and **VECTOR** without claiming authority over Army pipelines, tenure scrape, or manuscript prose.

What it is not

- **Not** the master plan itself.
- **Not** a claim that generative simulation **replicates** the empirical inverted-U on the same conditioning axis as 530 (`poolq_loo` / L_Q LOO).
- **Not** a substitute for reading `sports/documents/SPORTS_DATA_GAMEPLAN.md`, `tier_1_roadmap.md`, or `538_Cell10_Generative_Manual.md` when implementing or debugging notebooks.

1. Thesis frame (why SCOUT exists in this repo)

Charles's dissertation tests **advancement under constrained distinction**: individuals compete in **local talent pools** while advancement is **globally scarce**. Mid-quality peer environments can produce the highest individual advancement rates; elite environments create **congestion / signal compression** that drives rates back down (inverted-U).

Setting	Agent	Repo home	Outcome	Pool-quality construct
1 — Army	CODA	<code>talent/</code>	Promotion to Major vs attrition	Senior-rater LOO pool quality; CIF / Cox
2 — Basketball	SCOUT	<code>sports/</code>	NBA draft (<code>Y_draft</code>)	<code>poolq_loo</code> (LOO teammate <code>perf</code> , usually PPM z)
3 — Academia	PEER	<code>tenure/</code>	Tenure vs pre-tenure exit	<code>poolq_loo_mean</code> (LOO dept assistant pubs)

SCOUT’s job: Build and maintain the college basketball → NBA draft empirical stack (530 conductor, panel exports, forensics), the Alex-ordered Tier 1 modeling ladder (538 / 538D), and the modular generative lab (`tier1_pool_assignment.py`, CELL 10 playground) that stress-tests the minimal congestion score without over-claiming mechanism closure.

Why basketball is “Setting 2” in the cross-domain story: Global selection (draft) vs local opportunity (minutes, roles, teammate quality) is the cleanest measurement lab for Tier 1 objects (A, L_Q, L_C, Λ). Army is the anchor replication; academia is the third replication (preliminary per PEER report).

Cross-setting manuscript doc (VECTOR + Charles, June 2026):

`1-Various_PDE_and_Chat_stuff/5-Manuscript/advancement_under_constrained_distinction_dakota_feedback_v03.rtf` — committee-facing whole-project summary. Basketball is §1–§2 empirical anchor; §5–§7 generative framing aligns with SCOUT’s June status.

2. Current status — basketball / SCOUT (high level)

Established / working

Layer	Status	Primary artifacts / location
Panel conductor	✓	sports/530_sports_pipeline.ipynb → datasets/mbb/player_season_panel_530.csv
Empirical inverted-U	✓	Binned draft rate vs poolq_loo ; exports under datasets/mbb/exports_inverted_u_v0/ (e.g. 2026-06-02 run)
Wang empirical ladder	✓	sports/538_alex_tier1_model_and_fit.ipynb — CELLS 4–6 (bins → LPM → logit, L^*)
Pool forensics (PD11 targets)	✓	530 CELLS 5–9 — overlap, mean×SD, draftee-only variants
Tier 1 mechanism columns	✓	sports_pipeline/tier1_mechanism_vars.py — congestion_quality , crowding variants
Soft generative assignment	✓	sports/tier1_pool_assignment.py — $\tau \approx 0.65$ default, Plot A overlap calibration
Congestion selection score	✓	$S_i = A_i - w \cdot L_C$ with crowding_smooth ; crowding_l_z_scale for z-scored ability
Generative playground UI	✓	sports/538D_development.ipynb CELL 10; tier1_cell10_playground_run.py
539 score preset (partial DGP)	✓	Green button / SELECTION_539_* in tier1_sim_config.py — score + [0,1] scales, not full 539 assignment/noise/threshold
Plot B axis toggle	✓	SHOW_PLOT_B_TEAM_MEAN — False = L_Q LOO (530); True = team_mean (539-style)
HPC parameter sweeps	✓ (infrastructure)	sports/outputs/simulation_sweeps/ — Rivanna/Mac faithful 538 sweep runbooks

Major scientific finding (June 2026 — COMPASS must not garble this)

Same congestion score, different Plot B x-axis → different curve shape.

Plot B conditioning	Typical shape with 539 selection preset	Interpretation
team_mean (<code>SHOW_PLOT_B_TEAM_MEAN=True</code>)	Inverted-U (rise, peak, right-tail dip)	Matches 539 notebook readout (success vs team quality)
L_Q LOO (<code>SHOW_PLOT_B_TEAM_MEAN=False</code>)	Mostly decreasing	Matches 530 empirical axis question — standouts in weak peer pools selected most

Honest claims for manuscript / briefings: -  Empirical inverted-U on **real** college rosters vs LOO `poolq_loo` — **replicated** (stylized fact, not mechanism proof). -  Minimal score **equation** implemented: advancement score ↑ in own ability, ↓ in local congestion. -  Generative **proof-of-concept**: inverted-U vs **team_mean** under congestion; **not** bin-for-bin match on L_Q with same top- K knobs. -  Do **not** say “538D decomposes 539” or “generative replicates 530 on the same axis without qualification.”

Authoritative narrative doc: `Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md`

Three-level Wang sequence (oral + manuscript spine)

Level	Content	SCOUT status
0 — Equation	$S_i = A_i - \lambda L_{C,i}$	Written + coded
1 — 539 bundled POC	Sort-chop, score noise ϵ , 90th percentile threshold, plot vs team_mean	<code>539_alex_model.ipynb</code> — reference DGP, not architecture parent
2 — 538D modular lab	Soft assign $\tau \approx 0.65$, top- K , no score noise on main path, plot axis swappable	Active development notebook

Not done / open (sports-specific)

Item	Notes
Generative match to empirical LOO pool quality inverted-U	Deferred (Charles Path A) — parallel north-star; may need assignment noise, B term, or selection-rule work
538 empirical CELL 7+	Robustness (FE, clustering) — placeholder
535 roadmap TODOs	Cleaner CELL 5 regression block; theory language distinguishing <code>poolq_loo</code> vs mechanism columns
CELL 4D heterogeneity	Wired in 538D (<code>HETEROGENEITY_TOP_TAIL</code>); narrative parked until scaffolding clearer
537 legacy sim	Frozen — sort-and-chop; use only for Thread C promotion-score experiments or benchmark overlay
SR / BPM coverage gaps	Small colleges, 404 slugs — documented in <code>Pertinent_Thoughts_Scout.md</code>
Draft matcher 2c	Human label loop in gameplan still “planned”
Formal master plan	Deferred until Charles assigns COMPASS

3. Notebook and module architecture (SCOUT conductor map)

530_sports_pipeline.ipynb - DATA: spine → draft match → panel → ventile EDA / LPM
535_sports_tier_1.ipynb - MECHANICAL EDA: tier1_mechanism_vars, CELL 4 plots (roadmap contract)
538_alex_tier1_model_and_fit.ipynb - EMPIRICAL WANG LADDER + generative CELL 10–12 (Alex spine)
538D_development.ipynb - PRIMARY June 2026 generative lab (+ 4D heterogeneity)
537_Sports_Simulation.ipynb - FROZEN legacy (sort-and-chop, old Cell 10)
539_alex_model.ipynb - Bundled generative POC (one-night); reference only for score preset

File	Role
<code>sports/tier1_pool_assignment.py</code>	<code>soft_assign</code> , selection, <code>crowding_smooth</code> , scale helpers
<code>sports/tier1_sim_config.py</code>	Default constants, <code>SELECTION_539_*</code> , <code>SHOW_PLOT_B_TEAM_MEAN</code>
<code>sports/tier1_cell10_playground_run.py</code>	ipywidgets UI
<code>sports/tier1_generative_eda.py</code>	Plot B bin tables + figures
<code>sports/tier1_cell12_generative_eda.py</code>	Static replay from <code>tier1_cell10_playground_state.json</code>
<code>sports/tier1_heterogeneity_ventiles.py</code>	CELL 4D exports
<code>sports/tier1_539_reference_settings.json</code>	Reference λ , θ , γ for 539 preset

Package layer (`sports_pipeline/` — empirical panel)

Module	Role
<code>panel_build.py</code>	Panel construction, LOO recompute
<code>tier1_mechanism_vars.py</code>	<code>congestion_quality</code> , crowding columns
<code>paths.py</code>	<code>panel_530_csv()</code> → workspace <code>datasets/mbb/</code>

Workspace root panel path: `datasets/mbb/player_season_panel_530.csv` (not under `sports/datasets/`).

4. Key design decisions (basketball-specific)

- Realized pools vs generative rules are separate layers — 538 §6 measures pools the world formed; CELL 10 simulates assignment rules calibrated to 530 forensics (`Tier1_Presorting_Design_Note.md`).
- 539 vs 538D is parallel, not parent/child — 539 is a bundled proof-of-concept DGP; 538D is a modular lab anchored to 530 empirics. The **minimal claim** is the **score equation**, not either notebook.
- Plot axis is part of the estimand — Conditioning success on `poolq_loo` (LOO peer quality) vs `team_mean` is not interchangeable; June 2026 finding is a **feature** for distinguishing mechanism tests.
- τ calibration — Assignment temperature $\tau \approx 0.65$ targets 530 CELL 8 overlap (peak $\gg 1$, median roster SD ~ 0.8 z). Engineering detail for SCOUT; not centerpiece for VECTOR oral brief.
- Ability scale + congestion scale — When ability is z-scored, `crowding_1_z_scale` (p90–p10 of A) prevents crowding penalty from being negligible in selection.

6. **537 frozen** — Do not extend sort-and-chop for Thread A; red dashed benchmark overlay in Plot A only.
7. **Document-first split** — `SPORTS_DATA_GAMEPLAN.md` = strategic intent; `tier_1_roadmap.md` = 535 execution contract;
`Alex_Tier1_Sequential_Model_Outline.md` = advisor spine (5-Manuscript).
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5. Documentation inventory (SCOUT-authored or SCOUT-maintained, updated June 2026)

sports/documents/ — primary SCOUT contracts

Document	Role	Last refreshed
SPORTS_DATA_GAMEPLAN.md	Forward-looking pipeline contract; fast-path  ; Stage 5 Tier 1/generative	2026-06-08
tier_1_roadmap.md	535  contract, CELL 0, 538/538D generative block	2026-06-08
Tier1_Presorting_Design_Note.md	PD11 threads A/B/C; 530 calibration targets	2026-06-08
538_Cell10_Generative_Manual.md	CELL 10 operator manual	2026-06-08
Pertinent_Thoughts_Scout.md	Scratch discoveries + June generative section	2026-06-08
Alex_model_interpreted.md	PD12 interpretation; viable-peer signals table	2026-06-08
RIVANNA_RUNBOOK.md	HPC workflow	(ops; verify before large sweeps)
Mac_Faithful_538_Sweep_For_Dummies.md	Local sweep	
Rivanna_Faithful_538_Sweep_For_Dummies.md	Rivanna sweep	
537_Manual.md	Legacy sim only	
One_Page_Advisor_Brief_Template_College_Replication.md	Fill-in template (not auto-updated)	

Document	Role	Notes
Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md	Canonical June ground truth for generative + briefing	Lead doc for VECTOR
Scout_Modeling_Status_for_Vector_Barabasi_Briefing.md	May brief	Superseded banner → June doc
Alex_Tier1_Sequential_Model_Outline.md	Advisor-ordered spine §6A/ §7/§10	Updated 2026-06-08
Vector_to_Scout_Tier1_Modeling_Direction.md	VECTOR→SCOUT sync	June addendum in Final Note
alex_gates_briefing_structure_outline_v4.md	5-minute oral structure	§5 generative status June
important_stuff.md	Charles scratch anchor lines	
Tier1_Briefing_Outline.md	Equations + data map	Theory; not execution contract
Tier1_Narrative_Outline.md	Prose / competing forces	
tier_1_model.md	Model layer reference	

Cross-domain docs SCOUT reads but does not own

Document	Agent
talent/documents/Coda_Summary_For_Scout_and_Vector_Post_Replication.md	CODA
talent/documents/Army_to_College_Basketball_Replication_Handoff.md	CODA → SCOUT replication design
tenure/documents/TENURE_DATA_GAMEPLAN.md	PEER
PEER_Status_Update_for_VECTOR_2026-06-03.md	PEER

Stale / low-priority for COMPASS (unless scope expands)

- sports/documents/sports_mechanisms/advisor_packet*.md — older replication packets; may predate 538D.
- sports/documents/<!-- Generated by SpecStory, Markdown v2.md — chat export artifact; not a contract.
- Multiple alex_gates_briefing_structure_outline_v[1-3].md — superseded by v4 for oral flow.

6. Sister agents — what SCOUT knows (read their reports; don't merge pipelines)

Agent	Report	Maturity vs SCOUT	SCOUT dependency
CODA	<code>CODA_report_to_COMPASS.md</code>	Most mature empirical + survival stack	Inverted-U origin ; CIF/competing-risks vocabulary
PEER	<code>PEER_report_to_COMPASS.md</code>	Preliminary third setting (stage 9 bins)	Parallel LOO pool logic; tenure vs attrition
VECTOR	<i>(pending)</i>	Manuscript / theory	Consumes SCOUT status docs; does not run notebooks

SCOUT does not: merge Army OER snapshots into basketball panel, or tenure Wayback HTML into 530, or write dissertation chapter prose (VECTOR's lane).

7. Harmonization notes for COMPASS (language only — not code merge)

Concept	Army (CODA)	Basketball (SCOUT)	Academia (PEER)
Pool	Senior-rater cohort	Teammates (team-season)	Dept assistant cohort
LOO peer quality	Pool mean minus self	<code>poolq_loo</code>	<code>poolq_loo_mean</code>
Outcome	Promotion / separation	<code>Y_draft</code>	<code>tenure_event</code> / <code>attrition</code>
Inverted-U evidence	Established CIF panels	Established ventiles	Preliminary 18-bin stage 9
Generative layer	Less central in CODA report	Active (538D CELL 10)	Not SCOUT's scope
Competing risks framing	Native (Cell 11)	Draft yes/no (no time-to-event in ventile EDA)	Cox planned (Layer B; Cells 10/10.5/12 — not in <code>540</code> yet)

Shared mechanism phrase (VECTOR/Dakota): “Advancement under constrained distinction” — developmental benefits vs competitive constraints; finite distinction capacity in local comparison environments.

Wang ordering (cross-domain): (1) stylized fact, (2) minimal mechanism, (3) non-obvious predictions. SCOUT is between (2) and (3) on basketball generative axis.

8. Prior plan-map work (Cursor artifact — not master plan)

Charles and SCOUT previously mapped the doc ecosystem in `.cursor/plans/tier1_plan_map_c0d22e58.plan.md`. Key conclusion: **no single master plan file existed** as of June 2026; Dakota v03 RTF is the best **committee-facing whole-project narrative**; `Alex_Tier1_Sequential_Model_Outline.md` is the best **modeling spine** for basketball Tier 1.

COMPASS may absorb that map or supersede it — SCOUT does not treat the `.cursor/plans/` file as canonical repo documentation.

9. Clarifying questions for Charles — COMPASS update (2026-06-11)

COMPASS assigned. Charles locked several items; SCOUT still needs answers on open rows.

A. COMPASS process — resolved

#	Topic	Status
1–2	Single COMPASS agent; near-term = core manuscript	Done — <code>PROJECT_STATUS_AND_NEAR_TERM_PLAN.md</code>
3	Deadline	Summer–Fall 2026 draft/submission
4	VECTOR handoff	COMPASS reads <code>5-Manuscript/</code> + agent reports

B. Basketball generative priorities

- 5. **Resolved — manuscript-first (e)**: exports + axis table + nesting chain; LOO generative match **deferred**. Detail: `20260611_1626_COMPASS_to_SCOUT_model_coherence_questions.md`.
- 6. Should COMPASS treat **539** as deprecated, reference-only, or still active for Alex demos?
- 7. Is **team_mean** vs L_Q axis choice a **manuscript figure requirement** (show both) or internal lab detail?

C. Empirical basketball

- 8. Is `535` still an active notebook path, or is all Tier 1 empirical work consolidating in **538/538D**?
- 9. `PERF_METRIC` **lock**: Is PPM+z within-season still the canonical spec for cross-domain comparability, or should BPM/OBPM become primary before master plan freezes?

D. Cross-agent coordination

- 10. Should PEER adopt SCOUT’s **generative modular stack** for academia simulation (future), or keep PEER empirical-only until tenure Cox results land?
- 11. Does CODA’s **competing-risks** framing need a basketball analog (e.g. draft vs undrafted vs early exit) in the master plan, or is ventile draft-rate sufficient for Setting 2?

E. VECTOR / manuscript

12. Who **owns** updating Dakota RTF when PEER/CODA/SCOUT status shifts — VECTOR, Charles, or implementation agents?
 13. Should the master plan index `advancement_under_constrained_distinction_dakota_feedback_v03.rtf` as the external-facing spine, with internal gameplans as appendices?
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10. Suggested first tasks for COMPASS

Several items below are **done** or **in progress** (2026-06-11).

1. Read four agent reports + Dakota RTF + `Alex_Tier1_Sequential_Model_Outline.md`.
 2. Produce a **one-page index** doc in `3-Master_Plan/` linking: gameplans, status updates, primary notebooks, canonical outcomes per setting.
 3. Lock **claim language** table (what each setting can say in manuscript v1).
 4. Sequence remaining work: SCOUT nesting deliverables (blocking prose) vs PEER Cox (parallel) vs CODA TB-stratify (deferred unless elevated).
 5. Assign **single owner** per cross-cutting doc (who refreshes Dakota, Alex spine, gameplans).
 6. **Do not** merge `talent_pipeline/`, `sports_pipeline/`, and `tenure_pipeline/` codebases — harmonize at **estimand + narrative** layer only unless Charles explicitly requests shared Python abstractions.
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11. Agent identity note

SCOUT in Charles's naming = Cursor agent focused on `./sports/`, college basketball data, Tier 1 generative modules, and SCOUT-authored status docs in `5-Manuscript/`. SCOUT implements, documents, and stress-tests; **VECTOR** drafts manuscript prose; **Alex Gates** sets sequencing; **Charles** owns scientific claims and committee communication.

If a future Cursor session is labeled "SCOUT," it should read this report + `Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md` before changing generative defaults or briefing language.

SCOUT handoff report ends. COMPASS active — see `20260611_1626_COMPASS_to_SCOUT_model_coherence_questions.md` and `20260611_1626_COMPASS_to_SCOUT_questions.md`.